

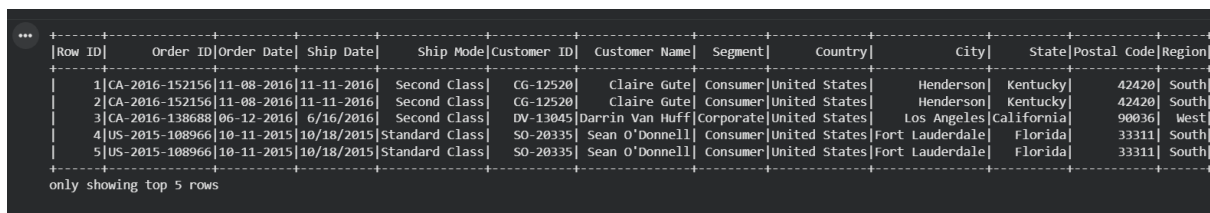
WEEK 7

Ques 1. Printing the first rows of the dataset using the head().

```
df = spark.read.csv(  
    "Superstore2.csv",  
    header=True,  
    inferSchema=True  
)
```

```
df.show(5)
```

Output:



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region
1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South
2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South
3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West
4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South
5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South

only showing top 5 rows

Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Col...	261.96	2	0	41.9136
FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric...	731.94	3	0	219.582
OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Add...	14.62	2	0	6.8714
FUR-TA-10000577	Furniture	Tables	Bretford CR4500 S...	957.5775	5	0.45	-383.031
OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Rol...	22.368	2	0.2	2.5164

Ques 2. Renaming the column

for col in df.columns:

```
df = df.withColumnRenamed(col, col.replace(" ", "_").replace("-", "_"))
```

```
print(df.columns)
```

Output:

```
['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode', 'Customer_ID', 'Customer_Name', 'Segment', 'Country', 'City', 'State', 'Postal_Code', 'Region', 'Product_ID',  
'Category', 'Sub_Category', 'Product_Name', 'Sales', 'Quantity', 'Discount', 'Profit']
```

Ques 3. Handling the missing values and removing the duplicate

```
df = df.na.fill({  
    "Sales": 0,  
    "Quantity": 0,  
    "Discount": 0,  
    "Profit": 0  
})
```

```
df = df.dropDuplicates()  
print("Rows:", df.count())
```

Output:

```
.. Rows : 9994
```

Ques 4. Saving the file as Delta and reading the delta file.

```
delta_path = "/content/delta/superstore"
```

```
df.write \  
    .format("delta") \  
    .mode("overwrite") \  
    .save(delta_path)
```

```
delta_df = spark.read.format("delta").load("/content/delta/superstore")  
delta_df.show(5, truncate=False)
```

Output:

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer
205	CA-2017-126382	06-03-2017	06-07-2017	Standard Class	HK-14890	Heather Kirkland	Corporate
233	US-2017-100930	04-07-2017	04-12-2017	Standard Class	CS-12400	Christopher Schild	Home Office
484	CA-2014-127691	7/22/2014	7/27/2014	Standard Class	EM-14065	Erin Mull	Consumer
531	US-2015-138303	09-03-2015	09-07-2015	Standard Class	MG-18145	Mike Gockenbach	Consumer

Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category
United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases
United States	Franklin	Tennessee	37064	South	FUR-FU-10002960	Furniture	Furnishings
United States	Tampa	Florida	33614	South	FUR-TA-10003473	Furniture	Tables
United States	New York City	New York	10024	East	TEC-AC-10002567	Technology	Accessories
United States	Philadelphia	Pennsylvania	19134	East	OFF-EN-10001335	Office Supplies	Envelopes

Product_Name	Sales	Quantity	Discount	Profit
Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	3083.43	7	0.5	-1665.0522
Eldon 200 Class Desk Accessories, Burgundy	35.168	7	0.2	9.6712
Bretford Rectangular Conference Table Tops	620.6145	3	0.45	-248.2458
Logitech G602 Wireless Gaming Mouse	159.98	2	0	57.5928
White Business Envelopes with Contemporary Seam, Recycled White Business Envelopes	52.512	6	0.2	19.692

Ques 5. Create a Incremental Dataset

```
incremental_data = [
    ("CA-2016-152156", "Second Class", 350.0, 6, 45.0),
    ("CA-2018-100001", "Standard Class", 180.0, 2, 25.0),
    ("CA-2018-100002", "First Class", 500.0, 5, 70.0)
]
```

```
columns = [
    "Order_ID",
    "Ship_Mode",
    "Sales",
    "Quantity",
    "Profit"
]
```

```
incremental_df = spark.createDataFrame(incremental_data, columns)
```

```
incremental_df.show()
```

Output :-

Order_ID	Ship_Mode	Sales	Quantity	Profit
CA-2016-152156	Second Class	350.0	6	45.0
CA-2018-100001	Standard Class	180.0	2	25.0
CA-2018-100002	First Class	500.0	5	70.0

Ques 6. Performing the MERGE operation.

```
from delta.tables import DeltaTable

# Load Delta Table

deltaTable = DeltaTable.forPath(spark, "/content/delta/superstore")

# Perform MERGE

(
    deltaTable.alias("old")
    .merge(
        incremental_df.alias("new"),
        "old.Order_ID = new.Order_ID"
    )
    .whenMatchedUpdate(set={
        "Ship_Mode": "new.Ship_Mode",
        "Sales": "new.Sales",
        "Quantity": "new.Quantity",
        "Profit": "new.Profit"
    })
    .whenNotMatchedInsert(values={
        "Order_ID": "new.Order_ID",
        "Ship_Mode": "new.Ship_Mode",
        "Sales": "new.Sales",
        "Quantity": "new.Quantity",
        "Profit": "new.Profit"
    })
)
```

```

        .execute()
    )

# Read the updated Delta table

merged_df = spark.read.format("delta").load("/content/delta/superstore")

print("Merged Delta Table:")

merged_df.show(20, truncate=False)

```

Output:

Merged Delta Table:

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment
2718	CA-2014-100006	09-07-2014	9/13/2014	Standard Class	DK-13375	Dennis Kane	Consumer
6288	CA-2014-100090	07-08-2014	07-12-2014	Standard Class	EB-13705	Ed Braxton	Corporate
6289	CA-2014-100090	07-08-2014	07-12-2014	Standard Class	EB-13705	Ed Braxton	Corporate
9515	CA-2014-100293	3/14/2014	3/18/2014	Standard Class	NF-18475	Neil Franz	Home Office
3084	CA-2014-100328	1/28/2014	02-03-2014	Standard Class	JC-15340	Jasper Cacioppo	Consumer
3836	CA-2014-100363	04-08-2014	4/15/2014	Standard Class	JM-15655	Jim Mitchum	Corporate
3837	CA-2014-100363	04-08-2014	4/15/2014	Standard Class	JM-15655	Jim Mitchum	Corporate
9441	CA-2014-100391	5/25/2014	5/29/2014	Standard Class	BW-11065	Barry Weirich	Consumer
6571	CA-2014-100678	4/18/2014	4/22/2014	Standard Class	KM-16720	Kunst Miller	Consumer
6570	CA-2014-100678	4/18/2014	4/22/2014	Standard Class	KM-16720	Kunst Miller	Consumer
6569	CA-2014-100678	4/18/2014	4/22/2014	Standard Class	KM-16720	Kunst Miller	Consumer
6572	CA-2014-100678	4/18/2014	4/22/2014	Standard Class	KM-16720	Kunst Miller	Consumer
9021	CA-2014-100706	12/16/2014	12/18/2014	Second Class	LE-16810	Laurel Elliston	Consumer
9020	CA-2014-100706	12/16/2014	12/18/2014	Second Class	LE-16810	Laurel Elliston	Consumer
6315	CA-2014-100762	11/24/2014	11/29/2014	Standard Class	NG-18355	Nat Gilpin	Corporate
6316	CA-2014-100762	11/24/2014	11/29/2014	Standard Class	NG-18355	Nat Gilpin	Corporate
6317	CA-2014-100762	11/24/2014	11/29/2014	Standard Class	NG-18355	Nat Gilpin	Corporate
6318	CA-2014-100762	11/24/2014	11/29/2014	Standard Class	NG-18355	Nat Gilpin	Corporate
9660	CA-2014-100860	3/26/2014	3/30/2014	Second Class	CS-12505	Cindy Stewart	Consumer
9462	CA-2014-100867	10/19/2014	10/24/2014	Standard Class	EH-14125	Eugene Hildebrand	Home Office

only showing top 20 rows

Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category
United States	New York City	New York	10024	East	TEC-PH-10002075	Technology	Phones
United States	San Francisco	California	94122	West	FUR-TA-10003715	Furniture	Tables
United States	San Francisco	California	94122	West	OFF-BI-10001597	Office Supplies	Binders
United States	Jacksonville	Florida	32216	South	OFF-PA-10000176	Office Supplies	Paper
United States	New York City	New York	10024	East	OFF-BI-10000343	Office Supplies	Binders
United States	Glendale	Arizona	85301	West	OFF-FA-10000611	Office Supplies	Fasteners
United States	Glendale	Arizona	85301	West	OFF-PA-10004733	Office Supplies	Paper
United States	New York City	New York	10035	East	OFF-PA-10001471	Office Supplies	Paper
United States	Houston	Texas	77095	Central	OFF-EN-10000056	Office Supplies	Envelopes
United States	Houston	Texas	77095	Central	FUR-CH-10002602	Furniture	Chairs
United States	Houston	Texas	77095	Central	OFF-AR-10001868	Office Supplies	Art
United States	Houston	Texas	77095	Central	TEC-AC-10000474	Technology	Accessories
United States	Springfield	Virginia	22153	South	FUR-FU-10002268	Furniture	Furnishings
United States	Springfield	Virginia	22153	South	TEC-AC-10001314	Technology	Accessories
United States	Jackson	Michigan	49201	Central	OFF-AR-10000380	Office Supplies	Art
United States	Jackson	Michigan	49201	Central	OFF-LA-10003930	Office Supplies	Labels
United States	Jackson	Michigan	49201	Central	OFF-PA-10001815	Office Supplies	Paper
United States	Jackson	Michigan	49201	Central	OFF-PA-10004082	Office Supplies	Paper
United States	Pomona	California	91767	West	OFF-LA-10001982	Office Supplies	Labels
United States	Lakewood	California	90712	West	TEC-PH-10004922	Technology	Phones

Product_Name	Sales	Quant	↑	↓	✎	🗑	⋮
AT&T EL51110 DECT	377.97	3		0			109.6111
Hon 2111 Invitation Series Corner Table	502.488	3		0.2			-87.9354
Wilson Jones Ledger-Size, Piano-Hinge Binder, 2" Blue"		196.704	6				0.2
Xerox 1887	91.056	6		0.2			31.8696
"Pressboard Covers with Storage Hooks, 9 1/2" x 11"	Light Blue"	3.928	1				0.2
Binder Clips by OIC	2.368	2		0.2			0.8288
Things To Do Today Spiral Book	19.008	3		0.2			6.8904
Strathmore Photo Frame Cards	14.62	2		0			6.7252
Cameo Buff Policy Envelopes	149.352	3		0.2			50.4063
DMI Arturo Collection Mission-style Design Wood Chair	317.058	3		0.3			-18.1176
Prang Dustless Chalk Sticks	2.688	2		0.2			1.008
Kensington Expert Mouse Optical USB Trackball for PC or Mac	227.976	3		0.2			28.497
Ultra Door Push Plate	29.46	6		0			9.7218
Case Logic 2.4GHz Wireless Keyboard	99.98	2		0			7.9984
Hunt PowerHouse Electric Pencil Sharpener, Blue	151.92	4		0			45.576
Dot Matrix Printer Tape Reel Labels, White, 5000/Box	196.62	2		0			96.3438
Xerox 1885	144.12	3		0			69.1776
Adams Telephone Message Book w/Frequently-Called Numbers Space, 400 Messages per Book	15.96	2		0			7.98
Smead Alpha-Z Color-Coded Name Labels First Letter Starter Set	18.75	5		0			9.0
RCA Visys Integrated PBX 8-Line Router	321.552	6		0.2			20.097

Ques 7. Validating the row count and checking the duplicates.

```
print("Final Row Count:", final_df.count())
```

Output:

```
... Final Row Count: 9996
```

```
from pyspark.sql.functions import col
```

```
duplicates = (
```

```
    final_df
```

```
    .groupBy("Order_ID")
```

```

        .count()
        .filter(col("count") > 1)
    )
    duplicates.show()
    print("Duplicate Count:", duplicates.count())

```

Output:

```

+-----+-----+
|      Order_ID|count|
+-----+-----+
|CA-2014-141838|    3|
|CA-2015-115798|    4|
|CA-2015-116750|    2|
|CA-2015-128083|    3|
|CA-2015-161830|    2|
|CA-2016-124016|    3|
|CA-2016-135776|    7|
|CA-2016-145730|    3|
|CA-2016-149783|    3|
|CA-2017-132521|    3|
|CA-2017-140326|    3|
|US-2017-111024|    3|
|US-2017-164147|    3|
|CA-2016-134936|    3|
|CA-2014-125612|    3|
|CA-2014-157546|    2|
|CA-2014-168592|    3|
|CA-2015-116484|    2|
|CA-2016-155474|    2|
|CA-2016-155978|    2|
+-----+-----+
only showing top 20 rows

Duplicate Count: 2471

```

Ques 8 . Summary after all the operation.

```
final_df.describe().show()
```

Output :-

summary	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment
count	9994	9996	9994	9994	9996	9994	9994	9994
mean	4997.5	NULL	NULL	NULL	NULL	NULL	NULL	NULL
stddev	2885.163629097427	NULL	NULL	NULL	NULL	NULL	NULL	NULL
min	1	CA-2014-100006	01-01-2017	01-01-2015	First Class	AA-10315	Aaron Bergman	Consumer
max	9994	US-2017-169551	9/30/2017	9/30/2017	Standard Class	ZD-21925	Zuschuss Donatelli	Home Office

Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name
9994	9994	9994	9994	9994	9994	9994	9994	9994
NULL	NULL	NULL	55190.3794276566	NULL	NULL	NULL	NULL	NULL
NULL	NULL	NULL	32063.69335036445	NULL	NULL	NULL	NULL	NULL
United States	Aberdeen	Alabama	1040	Central	FUR-BO-10000112	Furniture	Accessories	""While you Were...
United States	Yuma	Wyoming	99301	West	TEC-PH-10004977	Technology	Tables	netTALK DUO VoIP ...

Sales	Quantity	Discount	Profit
9996	9996	9994	9996
234.40964895832943	5.8288253809141946	0.31559491134928686	28.57454047619059
631.711752786377	25.518402688574024	3.3140086297924998	234.3583228517925
10/Pack"	1040 sheets"	30/Box"	-6599.978
999.98	98.352	98.352	8399.976

Ques 9 . Saving the updated file.

```
final_df.write \
    .format("delta") \
    .mode("overwrite") \
    .save("/content/delta/final_superstore")
```